

INTRODUCTION

Galaxy rotation curves remain a central test for theories of gravity and dark matter. Two paradigms dominate:

- **Λ CDM with NFW halos:** excellent per-galaxy fits with 4 free parameters (Υ_{disk} , Υ_{bulge} , M_{200} , c), but halo parameters vary widely across galaxies [7].
- **MOND:** a universal acceleration threshold $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$ with only 2 per-galaxy parameters (Υ_{disk} , Υ_{bulge}), but systematically poorer fits for some galaxy types [1, 6].

The Radial Acceleration Relation [RAR; 3, 4] demonstrates a tight empirical correlation between baryonic and observed accelerations, suggesting a universal transition between regimes. However, standard frameworks cannot simultaneously optimize for fit quality and parameter universality.

We ask: *does the hypothesis space contain families that occupy the trade-off region between NFW's fit quality and MOND's universality?*

DATA

We use the SPARC dataset [2]: 175 late-type galaxies with Spitzer $3.6 \mu\text{m}$ photometry and high-quality rotation curves. After filtering for quality flags $Q \leq 2$ and minimum 5 data points: 159 galaxies.

SPARC provides pre-computed velocity contributions (V_{disk} , V_{bulge} at $M/L = 1$, V_{gas}) at each radius, enabling mass model construction without analytic profile assumptions. The split is deterministic: 99 train, 30 validation, 30 test, stratified by luminosity.

For modified gravity laws, enclosed baryonic mass is derived as:

$$M_{\text{bary}}(r) = \frac{r}{G} (V_{\text{gas}}^2 + \Upsilon_d V_{\text{disk}}^2 + \Upsilon_b V_{\text{bul}}^2) \quad (1)$$

where $\Upsilon \in [0.1, 1.5]$ $M_{\odot}/L_{\odot}$ at $3.6 \mu\text{m}$.

METHOD

Law Families

Each family defines an acceleration function $a(r, M_{\text{enc}}, \rho)$. The three baselines are Newton baryons-only ($a = GM/r^2$, 2 params), Newton+NFW (standard halo, 4 params), and simple MOND ($a = a_N \nu(a_N/a_0)$, 2 params with a_0 fixed).

The generated family is **PIECEWISEREGIME**:

$$a(r) = a_0 \left[\sigma(x) \left(\frac{a_N}{a_0} \right)^{\alpha_{\text{in}}} + (1 - \sigma(x)) \left(\frac{a_N}{a_0} \right)^{\alpha_{\text{out}}} \right] \quad (2)$$

where $a_N = GM_{\text{bary}}/r^2$, $\sigma(x) = (1 + e^{-x})^{-1}$, $x = \log_{10}(a_N/a_0)/w$. All terms inside brackets are dimensionless; a_0 provides acceleration units. This family nests both Newton ($\alpha_{\text{in}} = \alpha_{\text{out}} = 1$) and deep-MOND ($\alpha_{\text{in}} = 1$, $\alpha_{\text{out}} = 0.5$) as special cases.

Four-Dimensional Fitness

Each family is scored on four independent axes (higher = better):

F1 (fit quality): $= 1/(1 + \text{median } \chi_{\text{red}}^2)$.

F2 (universality): $= 1/(1 + \overline{\text{CV}}_j)$, where $\text{CV}_j = \text{IQR}(\theta_j)/|\text{median}(\theta_j)|$ for each law parameter θ_j . Families with no law parameters (Newton baryons, MOND) get $F2 = 1$.

F3 (simplicity): $= 1/(k_{\text{law}} + k_{\text{galaxy}} + S)$, where S counts mathematical operations.

F4 (RAR transfer): $= 1/(1 + \text{RMS } \log_{10}(g_{\text{pred}}/g_{\text{obs}}))$.

No weighted sum. **Pareto dominance** determines optimal families: A dominates B iff $A \geq B$ on all axes and $A > B$ on at least one.

Mutation Operators

M4 (Freezing): Fix one law parameter at its cross-galaxy median.

M2 (Correction): Add multiplicative term $(1 + c_1 R^{p_1})$ with dimensionless ratio R .

M3 (Composition): Blend two families via sigmoid transition.

All mutations preserve dimensional consistency. Population capped at 30 families; structural complexity $S \leq 15$.

RESULTS

Baseline Comparison

Under BIC, Newton+NFW wins (median BIC = 26.2, 43/99 wins). **PIECEWISEREGIME**'s best-fit law parameters converge to $\alpha_{\text{in}} = 1.05$, $\alpha_{\text{out}} = 0.52$, recovering MOND-like exponents without prior knowledge.

Pareto Front and the Gap

On the 4D fitness, all four baselines are Pareto-optimal but no family achieves $F1 \geq 0.35$ and $F2 \geq 0.70$ simultaneously—a gap between NFW’s fit quality and MOND’s universality.

After two generations of evolution, **PR-only- a_0 -free**—PIECEWISEREGIME with $\alpha_{\text{in}} = 1.05$, $\alpha_{\text{out}} = 0.52$, $w = 0.615$ frozen, only a_0 varying per galaxy—fills this gap:

TABLE I. Test set 4D fitness (30 galaxies)

Family	F1	F2	F3	F4
Newton baryons	0.023	1.000	0.500	0.738
Newton+NFW	0.520	0.479	0.200	0.907
MOND-simple	0.261	1.000	0.250	0.886
PiecewiseRegime	0.424	0.575	0.077	0.908
PR-only-a_0	0.389	0.915	0.100	0.893

PR-only- a_0 -free achieves $\sim 85\%$ of NFW’s fit quality with nearly double the universality ($F2 = 0.92$ vs. 0.48) and one fewer per-galaxy parameter (3 vs. 4).

What Determines a_0 ?

Correlation of per-galaxy $\log_{10} a_0$ against six galaxy properties (159 galaxies): the strongest correlator is V_{max} ($r = 0.36$, $p = 2.4 \times 10^{-6}$), explaining only 13% of variance. No property exceeds $r = 0.4$. The a_0 scatter is 0.33 dex, largely independent of observed properties.

DISCUSSION

The universal transition shape ($\alpha_{\text{in}} \approx 1.05$, $\alpha_{\text{out}} \approx 0.52$, $w \approx 0.62$) is consistent with known RAR phenomenology [3, 4]. The variable- a_0 question connects

to the debate between (author?) [8] and (author?) [5]. Our contribution is not to this debate but to the methodology.

Limitations. (1) Single dataset (SPARC); no lensing or cluster constraints. (2) Mass-to-light degeneracy, though equal for all families. (3) PR-only- a_0 -free is phenomenological, not derived from a field theory. (4) The a_0 variation may reflect observational systematics.

What we cannot claim: discovery of new physics, evidence for or against dark matter, or that a_0 is a varying fundamental constant.

CONCLUSION

A multi-objective evolution engine identifies a Pareto-optimal law family combining NFW-level fit quality with MOND-level universality on 159 SPARC galaxies. The transition shape is universal; only the scale varies. The contribution is the method—multi-objective Pareto comparison of hypothesis families—not the astrophysical phenomenology, which is consistent with existing literature.

Code and data: <https://github.com/SaulVanCode/galaxy-rotation-worlds>

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